

Detecting Human Rights Violations on Social Media Using Machine Learning

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Abstract

The present-day Russia-Ukraine military conflict has exposed the pivotal role of social media in enabling the transparent and unbridled sharing of information directly from the frontlines. In conflict zones where freedom of expression is constrained and information warfare is pervasive, social media has emerged as an indispensable lifeline. Anonymous social media platforms, as publicly available sources for disseminating war-related information, have the potential to serve as effective instruments for monitoring and documenting Human Rights Violations (HRV). Our research focuses on the analysis of data from Telegram¹, the leading social media platform for reading independent news in post-Soviet regions.² We gathered a dataset of posts sampled from 95 public Telegram channels that cover politics and war news, which we have utilized to identify potential occurrences of HRV. Employing a mBERT-based text classifier, we have conducted an analysis to detect any mentions of HRV in the Telegram data. Our final approach yielded an F_2 score of 0.71 for HRV detection, representing an improvement of 0.38 over the multilingual BERT base model. We release two datasets that contain Telegram posts: (1) large corpus with over 1.6 millions posts and (2) annotated at the sentence-level dataset to indicate HRVs. The Telegram posts are in the context of the Russia-Ukraine war. We posit that our findings hold significant implications for NGOs, governments, and researchers by providing a means to detect and document possible human rights violations.

Introduction

The Russia-Ukraine war has raised significant concerns about Human Rights Violations (HRVs), with reports of atrocities against civilians and soldiers. Early in the conflict, the Office of the United Nations High Commissioner for Human Rights (OHCHR) documented thousands of civilian casualties, emphasizing the urgent need to respect international human rights laws (Bachelet 2022; OHCHR 2022).

Human rights are foundational to global and national governance, ensuring dignity and equality. However, recent trends, such as the World Justice Project's Rule of

Law Index, show a troubling global decline in these protections, driven by authoritarianism and weakened accountability (World Justice Project 2022). In response, technology plays an increasingly important role in documenting and preventing HRVs, including tools for online reporting and social media content as evidence (Dawes 2022; Bachelet 2022).

This research focuses on detecting HRV mentions in Russian-language social media data using Natural Language Processing (NLP). Telegram, a platform central to war-related discourse, is chosen for its popularity and secure communication features, especially among Russian-speaking users. With Telegram's rapid growth and role in disseminating politically sensitive information, it offers a critical dataset for this study.

We introduce two datasets:

1. **Corpus:** 1.6 million Telegram posts in Russian and Ukrainian.
2. **Annotated Subset:** 3,076 posts (15,693 sentences), labeled at the sentence level for binary HRV detection.

Key Contributions:

- The first ML research detecting HRVs in the Russian language.
- A publicly released corpus of 1.6 million Russian and Ukrainian Telegram posts.
- A labeled dataset identifying HRV mentions in Russian and Ukrainian.

This work lays the groundwork for tools that support human rights monitoring and contribute to UN Sustainable Development Goals by promoting peaceful and just societies.

Literature Review

Human Rights Violations and Machine Learning

With the widespread use of social media, researchers have explored its potential for detecting human rights violations (HRVs). Previous studies have applied computational methods to identify HRVs, including image processing approaches (Kalliatakis et al. 2017) and text analysis using social media data. For example, Chen et al. (Chen and Neill 2015) used social media graphs to cluster HRV-related tweets, while Alhelbawy et al. (Alhelbawy et al. 2020) applied deep neural networks to classify Arabic tweets as human rights abuse (HRA) or non-HRA, achieving a precision

of 85% and an F1-score of 75%. Despite these advances, our work differs by leveraging modern transformer models and focusing on random data sampling rather than keyword-based extraction.

Related Tasks

Since HRV detection using NLP is a nascent area, we draw inspiration from related fields like hate speech detection. Studies such as Gitari et al. (2015) used sentiment analysis for hate speech classification, while Schmidt and Wiegand (2017) reviewed NLP techniques for hate speech, identifying features like bag-of-words, sentiment analysis, and multimodal inputs. These methodologies provide valuable insights for HRV detection.

Telegram Data in Research

Telegram, a widely used platform in the post-Soviet regions, serves as a cornerstone of this research. Existing studies have primarily analyzed Telegram data for extremist networks (Urman and Katz 2022; Rogers 2020) or its role in resistance movements (Akbari and Gabdulhakov 2019). However, none have focused on Russian-language Telegram posts or HRV detection, highlighting the novelty of our approach.

Russia-Ukraine Conflict and NLP

Efforts to analyze the Russia-Ukraine conflict have largely used English Twitter data, focusing on hate speech, sentiment analysis, or public opinion (Thapa et al. 2022; Mir 2023). To our knowledge, this study is the first to utilize Telegram data in Russian to detect HRVs related to the Russia-Ukraine war, offering a unique contribution to assist NGOs and governments in tracking HRVs and potential war crimes.

Data

For the present study, we aimed to identify instances of human rights violations (HRVs) in textual data from public Telegram channels covering the Russia-Ukraine war and political news. To achieve this goal, we first identified a total of 95 public channels covering the aforementioned topics. These channels encompassed a wide range of pro-Russian and pro-Ukrainian affiliations, ensuring a diverse range of perspectives.

When considering the identification of affiliation in various channels, it is pertinent to observe that a significant number of these channels overtly express their support through the language employed. Notably, certain symbols or characters are utilized to indicate allegiance, such as the prevalent use of 'Z' as a pro-Russian emblem or the inclusion of a Ukrainian flag emoji as an indication of pro-Ukrainian sentiment. These affiliations are frequently evident in the channels' titles or descriptions, enabling prompt determination of the ideological inclination embraced by the channel. Examples of Telegram posts with their English translations can be found in Table 2.

We collected all posts from each channel starting from the beginning of the war on February 23rd, 2022 until January 29th, 2024. This yielded a large corpus consisting of posts written in both Russian and Ukrainian languages. We provide descriptive data for the corpus in Table 1. We openly

Total Number of Channels	95
Channels Leaning to Russia	28
Channels Leaning to Ukraine	26
Channels with Unclear Affiliation	41
Average Number of Posts per Channel	24,300

Table 1: Corpus Description

release this unique corpus for scientific community use.³

Annotated Data

To obtain a representative sample of posts related to the Russia-Ukraine conflict, we randomly extracted 30 posts from each of the 95 public channels identified on Telegram. This approach was chosen to ensure the sample's representativeness.

We split the posts into sentences for our task utilizing NLTK's PunktSentenceTokenizer library for Russian.⁴ The decision to partition the posts into sentences was made with the intention of conducting a more nuanced analysis. Additionally, since certain posts may contain numerous HRVs, our aim was to effectively capture each instance. The average length of a sentence in our annotated dataset was 13 words, with a mode of 7 words.

For the labeling task we employed the services of four annotators who were native Russian speakers and possessed diverse political affiliations. The annotators were provided with instructions in the Russian language.

Each post in our dataset was associated with the following information: date and timestamp, post ID, channel ID, and message. Human annotators labeled the dataset using Label Studio⁵ software and Google spreadsheets.

The human annotators tasked with labeling the data were instructed to identify sentences that potentially mention one or more human rights violations (HRVs). Specifically, we restricted this project to the following types of HRVs: killing civilians, destruction of civil objects, rape, torture, execution, or mistreatment of prisoners. Out of The Universal Declaration of Human Rights,⁶ we narrowed down the list of HRV to just six in order to make the task more specific for the annotators. We chose these particular six as the most relevant to the war and, hence, more likely appearing in our dataset. (We grouped 'rape, torture, execution' as one label, and hence have four types of HRVs in total). The final labels were binary in nature: 1 indicates the presence of a possible HRV, while 0 indicates the absence of HRVs. The annotators were presented with the full posts, but labeled the data on a sentence level, thereby facilitating more detailed and nuanced future analysis. We provide the annotator guidelines (in Russian) in our Github repository.⁷

³https://github.com/PoliNemkova/HRV_detection_on_social_media

⁴https://github.com/Mottl/ru_punkt

⁵<https://labelstud.io/>

⁶<https://www.ohchr.org/en/what-are-human-rights/international-bill-human-rights>

⁷The detailed annotation guidelines are available at

Positive examples	Translation
По предварительным данным, жертвами обстрела стали 9 мирных жителей (8 погибли и 1 ранен)», — сообщает СЦКК ЛНР	According to preliminary data, 9 civilians became victims of the shelling (8 were killed and 1 wounded),” the LPR JCCC reports.
По его словам, в результате ночного ракетного удара по Запорожью пострадали многоквартирные дома и улицы частного сектора города.	According to him, as a result of the night missile attack on Zaporozhye, apartment buildings and streets of the private sector of the city were damaged.
Negative examples	Translation
ВСУ официально заявили об уничтожении 44 из 50 запущенных по Украине ракет.	The Armed Forces of Ukraine officially announced the destruction of 44 out of 50 missiles launched in Ukraine.
О какой победе... Неужели украинцы не понимают, что их используют и всем наплевать на их жизни?	About what a victory ... Do the Ukrainians really not understand that they are being used and that no one cares about their lives?

Table 2: Examples of Labeled Sentences: (HRV or ‘1’ and non-HRV or ‘0’)

Most of the core annotations were performed by two annotators and adjudicated by a third. Specifically, a first annotator pre-labeled all 15,693 sentences (3076 posts); 228 sentences were labeled as HRV (positive class) and 15,465 as non-HRV (negative class). Then three subsets of posts were extracted: (1) those containing at least one sentence pre-labeled as positive, (2) an equal-sized random sample of posts where all of the sentences had been pre-labeled negative, and (3) all additional posts containing at least one sentence labeled positive by a preliminary version of our classifier (details of our model are covered in Section *Methods*). These posts were presented to the two annotators in a random order and double annotated. All disagreements were adjudicated by a different analyst who is a Russian speaking political expert. All the other pre-labeled posts were left in the dataset with their negative (non-HRV) sentence labels. After adjudication, the number of positive examples changed to 273, and negatives to 15,410.

The Cohen’s Kappa score (Cohen 1968), which is commonly used to evaluate annotator agreement, was computed to assess the level of agreement between the two independent annotators (before adjudication). The obtained Cohen’s Kappa score is 0.44 and indicates moderate agreement according to Landis and Koch [1977]. This score suggests that the task of identifying instances of human rights violations

Iteration	Language Chain
1	Russian-Arabic-Japanese-Russian
2	Russian-Turkish-Farsi-Russian
3	Russian-Chinese-Polish-Russian
4	Russian-Hebrew-German-Russian
5	Russian-Thai-Greek-Russian

Table 3: Back Translation Language Chains

in social media posts is challenging, and different annotators may have varying perceptions regarding what constitutes such violations. We provide positive and negative labeled examples in Table 2.

The final size of this annotated dataset is 15,693 labeled sentences: 273 examples of HRV class and 15,420 of non-HRV labeled examples.

Data Partitioning

From the annotated dataset previously described we randomly partitioned 80% of the posts for use as a training set, while reserving 20% for a held-out test set. Descriptions of our annotated training and test datasets are provided in Table 4.

Training Data

As described earlier and illustrated in Figure 4, our training data is highly imbalanced with just around 2% of sentences being labeled as HRV. We attempted to mitigate this class imbalance by utilizing the following data augmentation techniques: Back Translation and ChatGPT Data generation.

Back Translation. We applied paraphrasing of the positive examples using translation chains such as Russian-Arabic-Japanese-Russian (Beddiar, Jahan, and Oussalah 2021). This approach yielded another 228 HRV training examples per chain. In total, we utilized 5 different translation chains, each involving two languages in addition to the Russian source language (see Table 3). This resulted in a total of 1,140 additional positive examples to augment the training dataset, bringing number of positive examples to be close to 10% of the training data.

Chat GPT Data Generation. Furthermore, in order to enhance the training dataset, we implemented a method of generating synthetic data through the utilization of ChatGPT,⁸ which is a sizeable language model capable of generating language autonomously. Our approach entailed prompting ChatGPT to generate sentences in the Russian language that depict violations of human rights. Through a process of prompt engineering, we selected two prompts that yielded the most appropriate data for our purposes.

The first prompt instructed ChatGPT to produce sentences in the Russian language that exemplify instances of human rights violations during the Russia-Ukraine conflict. It is noteworthy that the initial output generated by ChatGPT exhibited a bias towards violations committed exclusively by the Russian side. This bias was subsequently addressed by

⁸https://github.com/PoliNemkova/HRV_detection_on_social_media.

⁸<https://openai.com/blog/chatgpt>

Parameter	Train	Test
Total Number of Posts	2,466	610
Total Number of Sentences	12,554	3,139
Number of HRV Sentences	228	45
Number of non-HRV Sentences	12,326	3,094
Number of HRV Posts	90	25
Number of non-HRV Posts	2,376	585

Table 4: The Base Dataset Description

posing the question to Chat GPT, "Why are all the violations presented as being committed by Russia only?" This led to a more balanced output of data.

Prompt 2 instructed ChatGPT to act as a Telegram channel and generate posts, each comprising a single sentence, that describe anecdotes of human rights violations in the context of the Russia-Ukraine war in 2022. This scenario-based prompt produced examples that were more aligned with the intended objective. However, during the data retrieval phase utilizing ChatGPT, we observed the model's instability, whereby the same prompt could elicit either a refusal to respond due to ethical considerations or a range of responses with varying styles.

After we reviewed the examples, five of them was removed as being irrelevant. We generated a total of 510 positive examples with this approach and included them as a part of our training data.

Sets of Training Data. We ran our models on the following sets of training data:

- The Base Training Dataset (D1): 80% of the annotated data (2,466 posts, 12,554 sentences).
- Augmented Training Dataset (D2): The Base Training Dataset + Augmented Data using Back Translation (3,606 posts, 13,694 sentences).
- Dataset 3 (D3): The Base Training Dataset + ChatGPT data (13,064 sentences).
- Dataset 4 (D4): Augmented Training Dataset + ChatGPT data (14,429 sentences).

Testing Data

The testing dataset consisted of the remaining 20% of the annotated posts, resulting in a size of 610 posts or 3,139 sentences.

Methods

The aim of our study is to detect textual mentions of human rights violations (HRV) in Telegram posts. In an application, these full posts would be forwarded, with the detected mentions highlighted, to an analyst for review, archival and action. Therefore, our data is labeled at the sentence level, rather than post level, to allow the sentence-level mentions to be emphasized. Hence, the methods here focus on performing classification at the sentence level.⁹

⁹Similar F_1 scores were seen when either classifying at the post level or rolling the sentence-level classifications up to evaluate at

Keyword	Translation	Keyword	Translation
'результате'	'resulted'	'людей'	'people'
'граждан'	'civilians'	'Буча'	'Bucha'
'пытки'	'tortures'	'мирного'	'peaceful'
'города'	'city'	'погиб'	'died'
'дома'	'houses'	'жителей'	'inhabitants'
'мирных'	'of peaceful'	'разрушен'	'destroyed'
'российские'	'Russian'	'обстрела'	'shelling'
'ВСУ'		'Информации'	
'Окупанты'		'получили'	
'ранения'		'детей'	
'пострадали'		'Мариуполя'	
'Милостив'		'САУ'	
'удар'		'сообщает'	
'дтей'		'словам'	
'видео'		'человека'	

Table 5: Keywords

In addition, our emphasis is on identifying most potential references to HRV, at the cost of a higher false positive rate. Therefore, we prioritize evaluating our system's recall over precision. Accordingly, we have opted to employ the F_{beta} score, where $beta$ is set to 2 (F_2), as our primary evaluation metric (Rijsbergen 1979). This measure enables us to place greater emphasis on recall than precision in our evaluation. We also report F_1 scores.

Baseline

In order to evaluate how well this task can be performed using a straight forward method, we evaluated a keyword baseline approach.

We utilized the *yake* library¹⁰ to extract 30 keywords and presented both the keywords themselves and their corresponding English translations in a Table 5.

It is worth noting that we also considered the suitability of the *nlk*¹² and *spaCy*¹³ Python libraries for the purposes of automated keyword extraction. However, based on our experimentation with a Russian dataset, we observed that the *yake* library produced a more pertinent set of keywords. Consequently, we adopted the *yake* library as our chosen method for keyword extraction.

Subsequently, we designated all sentences that contained at least one of the keywords extracted from the list as representing human rights violation (HRV), whereas those lacking any such keywords were categorized as non-HRV.

Models

For our tasks, we chose to test transformers as they are the state-of-the-art models for textual data.¹⁴ First, we employed a base multilingual BERT model (**mBERT**) which is a pre-trained deep learning language model with 12 layers (Transformer blocks), a word embedding (hidden) size

the post level. However, post-level classification results have higher variance, since the number of positive test examples is roughly cut in half.

¹⁰<https://pypi.org/project/yake/>

¹²<https://www.nltk.org/>

¹³<https://spacy.io/>

¹⁴<https://gluebenchmark.com/leaderboard>

Approach and Training Data	P	R	F_1	F_2
<i>D1: Base Training Dataset</i>				
Keywords Based (baseline)	0.12	0.65	0.19	0.35
mBERT base	0.42	0.31	0.36	0.33
RuBERT	0.46	0.36	0.40	0.38
XLM-RoBERTa	0.38	0.33	0.36	0.34
<i>D2: Augmented Dataset (base + backtranslated)</i>				
mBERT base	0.47	0.82	0.60	0.71
RuBERT	0.45	0.80	0.58	0.69
XLM-RoBERTa	0.41	0.80	0.54	0.67
<i>D3: Base Training Dataset + ChatGPT data</i>				
mBERT base	0.45	0.40	0.42	0.41
RuBERT	0.54	0.31	0.39	0.34
XLM-RoBERTa	0.39	0.53	0.45	0.49
<i>D4: Base Training Dataset + ChatGPT data + Augmented Data</i>				
mBERT base	0.39	0.82	0.53	0.67
RuBERT	0.43	0.69	0.53	0.62
XLM-RoBERTa	0.35	0.87	0.50	0.67

Table 6: Test Set Performance: Breakdown per model and training dataset.

of 768, and 12 self-attention heads (110M total parameters) (Devlin et al. 2018). Specifically, we used the Huggingface bert-base-multilingual-uncased model.¹⁵

We also evaluated **XLM-RoBERTa**,¹⁶ a transformer model pre-trained on a large corpus in a self-supervised fashion on 2.5TB of filtered CommonCrawl data spanning 100 languages (Conneau et al. 2019).

Finally, we also tried **RuBERT**, a BERT-based model specifically trained on the Russian language.¹⁷ RuBERT is a monolingual transformer-based language model that was trained from multilingual BERT with the addition of extra Russian vocabulary tokens (Kuratov and Arkhipov 2019).

We conducted k -fold cross-validation (Kohavi 1995) with k set to 5 for hyper-parameter tuning and model validation. During the tuning phase, we selected the training dataset, model, number of training epochs, and the block fine-tuning strategy that exhibited the optimal performance. Following cross-validation on the training dataset, we have determined that a total of 5 epochs is optimal. Additionally, we elected to freeze the first six model blocks and fine-tune only the

final six blocks as a means of mitigating overfitting and enhancing model generalization.

Several parameters were not tuned in the experiment. Specifically, all models were trained utilizing a categorical cross-entropy loss function, the AdamW optimizer (Loshchilov and Hutter 2017) and a learning rate of $5e-5$. In light of the class imbalance observed in the dataset, the weights in the loss function were modified to 0.17 for the non-HRV class and 0.83 for the HRV class. Alternative weight configurations of [0.5, 0.5] and [0.2, 0.8] were explored. The most optimal results for the experiment were determined empirically to be associated with the reported combined weight configuration.

Results

As previously stated, our emphasis lies on recall, therefore, we have adopted the utilization of F_2 as a principal evaluation metric. The most proficient model in our study, namely mBERT trained on the Augmented Dataset, obtained an F_2 score of 0.71, with precision and recall rates of 0.47 and 0.82, respectively. We observed that the incorporation of back translation data resulted in an enhancement of 0.38 points in the performance of mBERT relative to training only on the Base Training Dataset. All the performance scores are provided in Table 6.

Discussion

Model Performance. All of the models we tried performed best with the Augmented Dataset (D2). We attribute that to the fact that there is a substantial class imbalance in the Base Training Dataset. The top performing model is multilingual BERT, while RuBERT consistently performed just slightly worse (dropping 0.02 to 0.07 in the F_2 score) with the exception of training on D1 where RuBERT scored 0.05 points higher. This is counter-intuitive because both models have the same BERT base architecture and RuBERT was specifically trained for Russian. The difference appears to be in the training data. One possible explanation is that, as a result of the multi-language translation chain incorporating languages from divergent groups, the final back-translated “Russian” language resembles these intermediary languages. This could result in a lower performance for the Russian-specific LM, RuBERT, while mBERT could be robust to this grammatical and stylistic variation given its multi-lingual pre-training corpus.

XLM-RoBERTa achieved the highest F_2 score when trained on the augmented dataset (D2) and (D4). Notably, compared to other models, XLM-RoBERTa demonstrated the most significant improvement in performance when trained on the D3 dataset. This observation implies that the inclusion of ChatGPT had the greatest impact on this particular model, resulting in a 15% increase in F_2 score.

F_2 score for all models evaluated. Notably, incorporating ChatGPT data resulted in an 8% increase in mBERT and a 15% increase in XLM-RoBERTa, compared to training solely on the Annotated Dataset. However, for RuBERT, the addition of ChatGPT resulted in a 4% decrease in performance.

¹⁵<https://huggingface.co/bert-base-multilingual-uncased>

¹⁶<https://huggingface.co/xlm-roberta-base>

¹⁷<https://huggingface.co/DeepPavlov/rubert-base-cased>

The study found that combining all training datasets (D4) did not result in any performance improvement for the models evaluated, compared to the results achieved with D2 alone. Furthermore, the precision of the models was significantly compromised.

Class Imbalance. It is important to acknowledge the challenge that arises from the imbalance in the training data set, which includes only a scant number of instances of HRV compared to the considerable number of non-HRV sentences. Despite this discrepancy, our model was able to attain an F_2 score of 0.71. Nonetheless, the recall value of 0.84 and the precision of 0.45 suggest that the model is relatively well able to identify relevant items whenever they are present. Nevertheless, only 45% of the extracted items are deemed relevant. We will address the trade-off between precision and recall in our study in subsequent paragraphs.

Our attempts to tackle this challenge through data augmentation and assigning specific weights in the loss function have led to marginal improvements. Hence, the class imbalance remains a persistent obstacle in our data. Also, we recognize that the biggest issue to tackle now is an sub-optimal precision.

Impact of Data Augmentation. The issue of class imbalances was partially mitigated by the acquisition of additional positive examples through back translation. As demonstrated in Table 6, performance of all the models was observed to improve upon the inclusion of augmented back translation data. Specifically, the mBERT trained solely on the Base Training Dataset yielded an F_2 score of 0.33, while augmenting the training data resulted in an increase in the F_2 score to 0.71. It is noteworthy that the use of the Augmented Dataset for training resulted in a recall improvement of more than twofold across all models.

The utilization of a ChatGPT approach for augmenting the training data yielded enhanced performance compared to training solely on the Base Training Dataset (D1). XLM-RoBERTa exhibited the most substantial improvement with a 15% increase in the F_2 metric, alongside a marginal enhancement in precision by 1%, and a considerable improvement in recall by 20%.

Despite the fact that in our study, back translation brought the primary benefit, we note that data augmentation via ChatGPT also improved the performance. Our study focused predominantly on the Russian language, which is a high-resource language that can be translated accurately and with ease. Nonetheless, for other languages that do not possess such resources, data augmentation via ChatGPT represents a viable alternative for improving results.

Recall vs. Precision. Our system’s primary objective is to identify the maximum number of instances of HRV, even at the expense of erroneously classifying some non-HRV items. As such, our focus is on maximizing recall, rather than precision. (With that understanding, we utilize the F_2 metric as our principal evaluation criterion to evaluate our system’s performance.)

Our analysis of the various combinations of models and training data has revealed notable performance disparities between the most effective options, such as mBERT and RuBERT on both the Augmented Training Dataset and the Base

Training Dataset. Specifically, we observed that the most effective model was mBERT trained on the Augmented Training Dataset, which yielded a high recall rate of 0.82, albeit at the cost of a lower precision of 0.47 (and an F_2 score of 0.71). In contrast, RuBERT trained on the Augmented Training Dataset exhibited slightly inferior performance, with an F_2 score of 0.69, despite an identical recall rate of 0.80, but with a lower precision of 0.45.

Remarkably, we observed the highest recall rate of 0.87 when training XLM-RoBERTa on the combined dataset (D4), while simultaneously maintaining an F_2 score of 0.67.

The Cohen’s Kappa score. Despite the moderate value of the Cohen’s Kappa score, it is crucial to underscore the significance and complexity of our undertaking. We remain optimistic that our datasets and preliminary systems will provide valuable groundwork for further research by other scholars.

Error Analysis. It appears that the easiest type of HRV for the system to detect is ‘killing of civilians’, as seen in the following example, ‘Вчера вечером вблизи своего дома в Северодонецке из-за вражеских обстрелов погибли мужчина и женщина, еще две женщины убиты российскими обстрелами в Лисичанске и Привилле.’ (‘Last night, a man and a woman were killed near their home in Severodonetsk due to enemy shelling, two more women were killed by Russian shelling in Lisichansk and Priville’).

The most common *false negative* error is associated with ‘Civil Objects Destruction’. Examples of this error are ‘Загарбники здійснили масований обстріл міста Слов’янськ Донецької області — у зону обстрілу потрапив місцевий ринок.’ (‘The invaders launched a massive shelling of the city of Slovyansk, Donetsk region — a local market fell into the shelling zone.’) and ‘Когда увидел своими глазами, ужаснулся: два десятка домов в хлам’ (‘When I saw with my own eyes, I was horrified: two dozen houses got destroyed to the trash’). These errors suggest that adding context into the model may help the model to learn better and avoid similar errors.

The prevalent *false positive* examples are related to death or other violence but not in a context of HRV. For example: ‘Тут можна найти и конституционную реформу 2020 года, и проблемы с независимостью судов, и нарушения права на жизнь в ходе вооруженных конфликтов, и пытки, и домашнее насилие’ (‘Here you can find the constitutional reform of 2020, and problems with the independence of the judiciary, and violations of the right to life during armed conflicts, and torture, and domestic violence’). This suggests that we need to provide more negative examples that include references to death yet are not considered as an HRV.

Possible Improvements. A potential avenue for future improvement could involve integrating contextual information from posts into the classifier. We posit that such an approach has the potential to enhance the performance of the classifier.

Future Work. With our encouraging results, we are eager to explore further improvements in our system. Our research findings offer the possibility of extending this project by developing a multiclass classifier capable of detecting specific

types of HRV. We are particularly interested in the four types of HRV mentioned in the annotation description, namely, (1) killing civilians, (2) destruction of civil objects, (3) rape, torture, and execution, and (4) mistreatment of prisoners.

We also explore applications of generative models, such as GPT, in the project and future publications.

We see our work as a first step in a project to build a more general social media based tool for documenting human rights violations during wars. Specifically, future work includes extrapolating this model to other languages and geographic locations.

Conclusion

Detecting and documenting human rights violations (HRVs) is crucial yet resource-intensive. Our study addresses this challenge by leveraging machine learning and natural language processing to develop cost-effective methods for HRV detection, focusing on data from the Russia-Ukraine conflict.

Social media platforms, particularly Telegram, are valuable sources of information but pose challenges due to high data volume, misinformation, and propaganda. To address this, we collected and annotated data from 95 public Telegram channels covering war-related news, applying data augmentation and transformer-based classification. Our best model, fine-tuned mBERT, achieved an F_2 score of 0.71.

This study is the first to use Russian-language data for HRV detection, contributing two publicly available datasets: (1) a large corpus of 1.6 million Telegram posts and (2) an annotated dataset labeled for HRVs. These resources aim to support further research on HRV detection in Eastern Europe.

We believe this work provides a foundation for NGOs, governments, and researchers to identify and document HRVs more effectively, promoting accountability and human rights protection.

Ethical Statement

Data Collection

The data collection process relied on the Telegram API.¹⁸ Joining (or subscribing) the public channels was not necessary in order to collect the data, though a publicly available invite link and a Telegram account was required in order to access the channel. (We provide channels links in the released dataset).

Initially, we conducted manual searches and utilized online resources to identify pertinent channels. Subsequently, the channel IDs were employed to extract the posts via the API, covering the time span from February 23, 2022, to January 29, 2024. We have included comprehensive instructions and a pre-configured code to facilitate data extraction through our prescribed method.¹⁹ Users do need to apply for a personal Telegram API key, which is free.

Each post is characterized by distinct identifiers, including the channel ID, post ID, date (comprising day, month, year, and time), and the textual content of the post.

Dataset Release Safety

We believe that the Telegram dataset holds significant value for the research community, particularly in the context of applying natural language processing (NLP) techniques to analyze the Russia-Ukraine conflict. This is evident from the growing interest of researchers worldwide, as reflected in the numerous scholarly papers that have utilized Twitter data as a primary source (such as Garcia and Cunanan-Yabut; Polyzos; Vahdat-Nejad et al.; Shlkamy, Mahar, and Sedky; Caprolu, Sadighian, and Di Pietro). However, it is important to note that Twitter has been predominantly utilized for gathering NLP datasets for social media analysis, while the relevance of Telegram, especially in Russian-speaking countries, has been comparatively understudied. As previously noted, Telegram exhibits a notably higher prevalence within the Russian-speaking region compared to Twitter. Consequently, we encourage researchers to expand their analyses to include Telegram, as this may yield more insightful and contextually relevant results.

Nevertheless, it is essential to acknowledge the potential risks associated with the usage of Telegram data, particularly in Russia, where individuals can face arrest for certain posts on the platform.²⁰ Additionally, there is a possibility that channels and posts may be sporadically deleted. To mitigate against our dataset inadvertently causing harm, we have adopted a cautious approach by only releasing the channel and post IDs, along with instructions for extracting them. By adopting this strategy, the dataset comprises solely publicly available channels and posts as of the date of the API call.

Although, to our best knowledge, there is currently no explicit policy from Telegram regarding the use of public data, we have followed a similar approach to Twitter’s policy. We provide a dataset that includes channel IDs and post IDs, which are sufficient for retrieving a post. Consequently, if a channel or post is subsequently deleted from the platform, we do not retain any information associated with it.

Additionally, it is worth noting that Telegram channels identified within the dataset do not allow user participation in the main channel, with the occasional exception of comments and emoji reactions, mitigating risk of individual identification on a member level.

Scientific Neutrality Statement

We emphasize that irrespective of the author’s personal perspectives, our objective is to construct an impartial system capable of capturing instances of human rights violations, regardless of the responsible parties involved.

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¹⁸<https://core.telegram.org/>

¹⁹https://github.com/PoliNemkova/HRV_detection_on_social_media

²⁰<https://www.washingtonpost.com/opinions/interactive/2023/political-protest-new-generation-faces/>

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References

- Akbari, A.; and Gabdulhakov, R. 2019. Platform surveillance and resistance in Iran and Russia: The case of Telegram. *Surveillance & Society*, 17(1/2): 223–231.
- Alhelbawy, A.; Lattimer, M.; Kruschwitz, U.; Fox, C.; and Poesio, M. 2020. An nlp-powered human rights monitoring platform. *Expert Systems with Applications*, 153: 113365.
- Bachelet, M. 2022. Ensuring accountability for atrocities committed in Ukraine. Last accessed 24 February 2023.
- Beddiar, D. R.; Jahan, M. S.; and Oussalah, M. 2021. Data expansion using back translation and paraphrasing for hate speech detection. *Online Social Networks and Media*, 24: 100153.
- Caprolu, M.; Sadighian, A.; and Di Pietro, R. 2022. Characterizing the 2022 Russo-Ukrainian Conflict Through the Lenses of Aspect-Based Sentiment Analysis: Dataset, Methodology, and Preliminary Findings. *arXiv preprint arXiv:2208.04903*.
- Chen, F.; and Neill, D. B. 2015. Human rights event detection from heterogeneous social media graphs. *Big Data*, 3(1): 34–40.
- Cohen, J. 1968. Weighted kappa: nominal scale agreement provision for scaled disagreement or partial credit. *Psychological bulletin*, 70(4): 213.
- Conneau, A.; Khandelwal, K.; Goyal, N.; Chaudhary, V.; Wenzek, G.; Guzmán, F.; Grave, E.; Ott, M.; Zettlemoyer, L.; and Stoyanov, V. 2019. Unsupervised cross-lingual representation learning at scale. *arXiv preprint arXiv:1911.02116*.
- Dawes, J. 2022. Autonomous weapons as a solution to war crimes? Last accessed 24 February 2023.
- Devlin, J.; Chang, M.-W.; Lee, K.; and Toutanova, K. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Garcia, M. B.; and Cunanan-Yabut, A. 2022. Public Sentiment and Emotion Analyses of Twitter Data on the 2022 Russian Invasion of Ukraine. In *2022 9th International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)*, 242–247. IEEE.
- Gitari, N. D.; Zuping, Z.; Damien, H.; and Long, J. 2015. A lexicon-based approach for hate speech detection. *International Journal of Multimedia and Ubiquitous Engineering*, 10(4): 215–230.
- Kalliatakis, G.; Ehsan, S.; Fasli, M.; Leonardis, A.; Gall, J.; and McDonald-Maier, K. D. 2017. Detection of human rights violations in images: Can convolutional neural networks help? *arXiv preprint arXiv:1703.04103*.
- Kohavi, R. 1995. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Ijcai*, volume 14, 1137–1145. Montreal, Canada.
- Kuratov, Y.; and Arkhipov, M. 2019. Adaptation of deep bidirectional multilingual transformers for russian language. *arXiv preprint arXiv:1905.07213*.
- Landis, J. R.; and Koch, G. G. 1977. The measurement of observer agreement for categorical data. *biometrics*, 159–174.
- Loshchilov, I.; and Hutter, F. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.
- Mir, A. A. 2023. Exploring the perceived opinion of social media users about the Ukraine–Russia conflict through the naturalistic observation of tweets. *Social Network Analysis and Mining*, 13(1): 44.
- OHCHR. 2022. SITUATION OF HUMAN RIGHTS IN UKRAINE IN THE CONTEXT OF THE ARMED ATTACK BY THE RUSSIAN FEDERATION. Last accessed 24 February 2023.
- Polyzos, E. 2022. Escalating tension and the war in Ukraine: Evidence using impulse response functions on economic indicators and twitter sentiment. *Available at SSRN 4058364*.
- Rijsbergen, C. 1979. Information retrieval 2nd ed Butterworth. *London [Google Scholar]*, 115.
- Rogers, R. 2020. Deplatforming: Following extreme Internet celebrities to Telegram and alternative social media. *European Journal of Communication*, 35(3): 213–229.
- Schmidt, A.; and Wiegand, M. 2017. A survey on hate speech detection using natural language processing. In *Proceedings of the fifth international workshop on natural language processing for social media*, 1–10.
- Shlkamy, E. S. I.; Mahar, K. M.; and Sedky, A. A. H. 2023. A Russia-Ukraine Conflict Tweets Sentiment Analysis Using Bidirectional LSTM Network.
- Thapa, S.; Shah, A.; Jafri, F.; Naseem, U.; and Razzak, I. 2022. A multi-modal dataset for hate speech detection on social media: Case-study of russia-ukraine conflict. In *Proceedings of the 5th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE)*, 1–6.
- Urman, A.; and Katz, S. 2022. What they do in the shadows: examining the far-right networks on Telegram. *Information, communication & society*, 25(7): 904–923.
- Vahdat-Nejad, H.; Akbari, M. G.; Salmani, F.; Azizi, F.; and Nili-Sani, H.-R. 2023. Russia-Ukraine war: Modeling and Clustering the Sentiments Trends of Various Countries. *arXiv preprint arXiv:2301.00604*.
- World Justice Project. 2022. World Justice Project Rule of Law Index® 2022 Insights. Last accessed 24 February 2023.